

Number Keypad

Operation

Keystroke

Function

Note

PEL Number Keypad Handling

o Show Numlock State

• Keys when not numlock

• Keys with numlock

• Toggle PEL Keypad numlock mode

o PEL copy keypad commands

• PEL kill keypad commands

• Implementation notes

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Testing the Keys from the shell

To see if a key generates an escape code sequence, run the following command from a shell and type the specific key(s). The escape sequences generated by key press can be shown in the terminal shell by one of these 2 methods:

• typing Control-V followed by the key.

• Executing: sed -n 1Where the last character is the L lower case letter. This prints all values in caret (^) format.

Show PEL Numlock Mode state

<f11> ? k #

(pel-show-mac-numlock)

Display state of PEL Keypad num-lock mode.

• This has the same limitation as some of the of the key bindings described below: in some cases, such as when Emacs runs in terminal mode under macOS Terminal, the numlock state of the keypad cannot be identified by Emacs.

Key behaviour when Number Keypad is not num-locked

See also:

• Navigation

• Cut & Paste

• Text Modes

• Scrolling

Special customization:

• pel-keypad-0-is-kp-yank

• pel-keypad-+-is-kp-add

Access their customization buffer via:

• <f11> = <f2>

• <f11> <f2> g pel-keypad-keys

See Customize

Not Numlocked

pel-toggle-mac-numlock	=	/	*
pel-home	forward-line -1	pel-scroll-down	pel-kill-or-delete-marked-or-whole-line
left-char	recenter-top-bottom	right-char	pel-copy-marked-or-whole-line
pel-end	forward-line	pel-scroll-up	<enter>
yank		delete-char	

With PEL, right after pel-init is called, the number keypad is set to non num-lock mode and 14 of the 18 keys take the special meaning described in the picture to the left.

<clear>

• With Emacs in graphics mode the top-left-most key registers as <clear> and is explicitly bound by PEL to pel-toggle-mac-numlock to switch the keypad numlock mode on or off.

• With Emacs in terminal mode running under Terminal.app with the settings identified in the terminal settings which maps the <clear> key to NumLock key.

• However, Emacs operates in terminal (TTY) mode, hitting <clear> changes the keypad key meaning but the <clear> key does not generate an event to inform Emacs. Because of that, in macOS terminal mode, the pel-toggle-mac-numlock command bound to the <f11> # key sequence does nothing and warns.

You can still hit the <clear> key in macOS terminal mode to toggle the NumLock mode on or off though and the number keypad keys will work appropriately.

Cursor keys:

• The 4 keys implement cursor functionality according to the normal cursor positions and implement the functions required for left, up, right and down.

The pel-home and pel-end commands are available in the left column.

The center key, the 5 key, is bound to recenter-top-bottom.

The pel-scroll-down & pel-scroll-up are available in the right column.

The “0” key is mapped to yank.

• In some situations, like when when using Emacs on a Linux host accessed through ssh, this may not work.

Try setting pel-keypad-0-is-kp-yank to t.

If that fails se the standard C-y instead.

The “.” key is bound to delete-char.

The “-” key is pel-kill-delete-marked-or-whole-line

The “+”key is pel-copy-marked-or-whole-line

With Emacs running in terminal mode, the pel-keypad-+-is-kp-add user option must often be set to t for this to work.

Key behaviour when Number Keypad is num-locked

Numlocked

pel-toggle-mac-numlock	=	/	*
7	8	9	-
4	5	6	+
1	2	3	<enter>
0		.	

Note that PEL activates non-numlock mode by default: to activate the numlock mode you can use the <f11> # key sequence or press the top-left-most key (in graphics mode only): this executes pel-toggle-mac-numlock.

Note:

With a terminal-mode Emacs running in Linux in a VM hosted on macOS, do not use <f11> # . Just use the <clear> key. Otherwise the internal state PEL sees may be out of sync and the keypad keys will not behave properly. In that case, try pressing <clear> again.

Toggle PEL Keypad Numlock mode

• <f11> #

• <clear>

(pel-toggle-mac-numlock)

Toggle PEL numlock mode for Emacs running in graphics mode.

• Complements the use of NumLock key (or <clear> used as NumLock).

This does not work when Emacs is running in macOS terminal mode. In that case the command simply print a warning message reminding you.

• When running in a terminal under Linux or running macOS under Terminal.app properly configured (as described in terminal settings), you can use the <clear> key to toggle from NumLock to non-NumLock mode but it will not execute the command.

To find out the state, press one of the keypad number keys to see if the result is insertion of a number or a cursor operation.

When running Linux in a VM hosted under macOS, PEL assumes invalidly assume Linux has access to a real NumLock key. In that case just do never use this command from <f11> # and always use the <clear> key.

With PC computers the top-left-most key is an explicit num-lock key.

See extra info related to macOS keyboard below.

Operation	Keystroke	Function	Note																			
PEL Copy Keypad Keys	The “+” keypad key can also be used for copy operation. The first of the 3 binding only works when PEL is in non numlock mode, but the other 2 bindings use modifier keys and the commands are bound regardless of the PEL numlock mode.																					
Copy region or line at point ★PEL Enhanced Key ★ Available in PEL non numlock mode See also: 🔗 Cut & Paste	• M-w • <f11> = 1 • <f11> + • <kp-separator>	(pel-copy-marked-or-whole-line)	Flexible copy to kill ring.: copy visible region if any, otherwise copy current line to kill ring. The copy operation is controlled by the (optional) argument: • If N = 0: copy region (regardless of whether it is visible or not. • If a region is active/visible: copy the region's text. • if no region is active/visible copy N lines: • If no argument, (N=1) copy current line. • If N > 0: copy current line and N-1 following lines. • If l < 0: copy current line and N-1 previous lines. All copied lines are complete. The copied text is saved in the kill-ring. All copy operations are performed by ‘kill-ring-save’ (the original binding for that key). • Replaces standard binding to kill-ring-save which only copies region. • In graphics mode: text is also copied to the OS clipboard. • In terminal (TTY) mode the keypad + key is interpreted as <kp-separator> on macOS so this key is bound to the command (in non numlock mode)																			
Copy complete word at point See also: • 🔗 Cut & Paste • 🔗 Text Modes	• <f11> = w • C-<kp-add>	(pel-copy-word-at-point)	Copy word at point. • Shows the text copied in the echo area. 👉 See table 🔗 Text Modes for information on text modes that affects this. • The <f11> t m ? command displays the mode and the <f11> t m prefix allows modifications of the mode. • See changing the word mode to include or exclude some characters as word delimiters: • subword-mode . To toggle that mode: <f11> t m b • superword-mode . To toggle that mode: <f11> t m p																			
Copy complete symbol at point See also: 🔗 Cut & Paste	• <f11> = . • M-+ • M-<kp-add>	(pel-copy-symbol-at-point)	Copy symbol at point. Syntax depends on the syntax table for the buffer. • Shows the text copied in the echo area. 👉 The syntax of the symbol depends on the major mode used by the current buffer.																			
PEL Kill Keypad Keys	The “-” keypad key can also be used for kill operation. The binding only works when PEL is in non numlock mode.																					
Kill/Delete marked region/line(s) ★PEL Enhanced Key ★ Available in PEL non numlock mode See also: 🔗 Cut & Paste	• C-w • <f11> - 1 • <kp-subtract> • ⌘-x	(pel-kill-or-delete-marked-or-whole-line &optional N)	Flexible region/whole-line kill/delete. • N=0 := kill region (active/visible or not) • Sign of N selects operation: • positive := kill (default) • negative := delete • Select text to delete/kill based on presence of region: • if a region is marked: kill/delete region's text, • if no region: kill/delete abs(N) lines, start at point. • If operation is to kill 1 line and the line is empty, then delete line instead of killing it.																			
✂️ This replaces the standard Emacs binding to kill-region which always kill text between mark and point, even when the region is not marked. When text is killed it is killed with kill-region, so it retains the filtering and kill ring text appending capabilities. 👉 In graphics mode this also copies text to the OS clipboard. 👉 With PEL in non-numlock mode, the “-” key on the number keypad is bound to this command. 🍎 On macOS in graphics mode only: PEL rebinds ⌘-x from (kill-region) to this command, making this easy to use key able to perform more.																						
Implementation Notes 🍎	On macOS keyboards with number keypads, the keys available when Emacs runs in graphics mode differ from the keys available when Emacs runs in terminal mode.																					
	🍎 “Emacs -Q” Keypad in Graphics mode <table><tr><td><clear></td><td>=</td><td><kp-divide></td><td><kp-multiply></td></tr><tr><td><kp-7></td><td><kp-8></td><td><kp-9></td><td><kp-subtract></td></tr><tr><td><kp-4></td><td><kp-5></td><td><kp-6></td><td><kp-add></td></tr><tr><td><kp-1></td><td><kp-2></td><td><kp-3></td><td rowspan="2"><kp-enter></td></tr><tr><td colspan="2"><kp-0></td><td><kp-decimal></td></tr></table>		<clear>	=	<kp-divide>	<kp-multiply>	<kp-7>	<kp-8>	<kp-9>	<kp-subtract>	<kp-4>	<kp-5>	<kp-6>	<kp-add>	<kp-1>	<kp-2>	<kp-3>	<kp-enter>	<kp-0>		<kp-decimal>	When Emacs is running in graphical mode, the <clear> key is available and the number keys are distinguishable from the self-insert digits. The keys on the right-most row are also distinguishable and so is the key labelled <kp-decimal>.
<clear>	=	<kp-divide>	<kp-multiply>																			
<kp-7>	<kp-8>	<kp-9>	<kp-subtract>																			
<kp-4>	<kp-5>	<kp-6>	<kp-add>																			
<kp-1>	<kp-2>	<kp-3>	<kp-enter>																			
<kp-0>		<kp-decimal>																				
	🍎 “Emacs -Q” Keypad in Terminal.app TTY mode <table><tr><td></td><td>=</td><td>/</td><td>*</td></tr><tr><td>7</td><td>8</td><td>9</td><td><kp-subtract> -</td></tr><tr><td>4</td><td>5</td><td>6</td><td><kp-separator> +</td></tr><tr><td>1</td><td>2</td><td>3</td><td rowspan="2">RET</td></tr><tr><td colspan="2">0</td><td>M-O n</td></tr></table>			=	/	*	7	8	9	<kp-subtract> -	4	5	6	<kp-separator> +	1	2	3	RET	0		M-O n	When Emacs is running in terminal (TTY) mode inside Terminal.app: • The <clear> key is not detectable. • The +, / and * keys only register as self-insert. The digit keys register as self-insert digits but if we bind the corresponding <kp-digit> key Emacs is able to handle it properly. • On the right-most row the <kp-subtract> is detectable, but the key below is detected as <kp-separator> instead of the normal <kp-add>. • Also, the <kp-decimal> is not detected, instead Emacs detects the key sequence M-O n.
	=	/	*																			
7	8	9	<kp-subtract> -																			
4	5	6	<kp-separator> +																			
1	2	3	RET																			
0		M-O n																				